

```
import cv2

img = cv2.imread("Ex1.jpg") # Corrected path to the image

if img is None:
    print("Error: Image 'Ex1.jpg' not found. Please make sure the image is uploaded to the Colab environment (e.g., to the /cor
else:
    gray_img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    cv2.imshow("Original Image", img)
    cv2.imshow("Grayscale Image", gray_img)

    cv2.imwrite("gray_image.jpg", gray_img)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```






```
import cv2
from google.colab.patches import cv2_imshow

image = cv2.imread("Ex2.jpg") # Corrected path for Colab environment

if image is None:
    print("Error: Image 'Ex2.jpg' not found. Please ensure the image is uploaded to the Colab environment (e.g., to the /content directory.)")
else:
    blur = cv2.GaussianBlur(image, (5,5), 10)

    cv2_imshow(image)
    cv2_imshow(blur)

    cv2.imwrite("blurred_image.jpg", blur)
```










```
import cv2
from google.colab.patches import cv2_imshow

img = cv2.imread("Ex3.jpg") # Corrected path for Colab environment

if img is None:
    print("Error: Image 'Ex3.jpg' not found. Please ensure the image is uploaded to the Colab environment (e.g., to the /cont
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    edges = cv2.Canny(gray, 100, 200)

    cv2_imshow(img)
    cv2_imshow(edges)

    cv2.imwrite("Canny_Edges.jpg", edges)
```




```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow

img = cv2.imread("Ex4.jpg", cv2.IMREAD_GRAYSCALE) # Corrected path for Colab environment

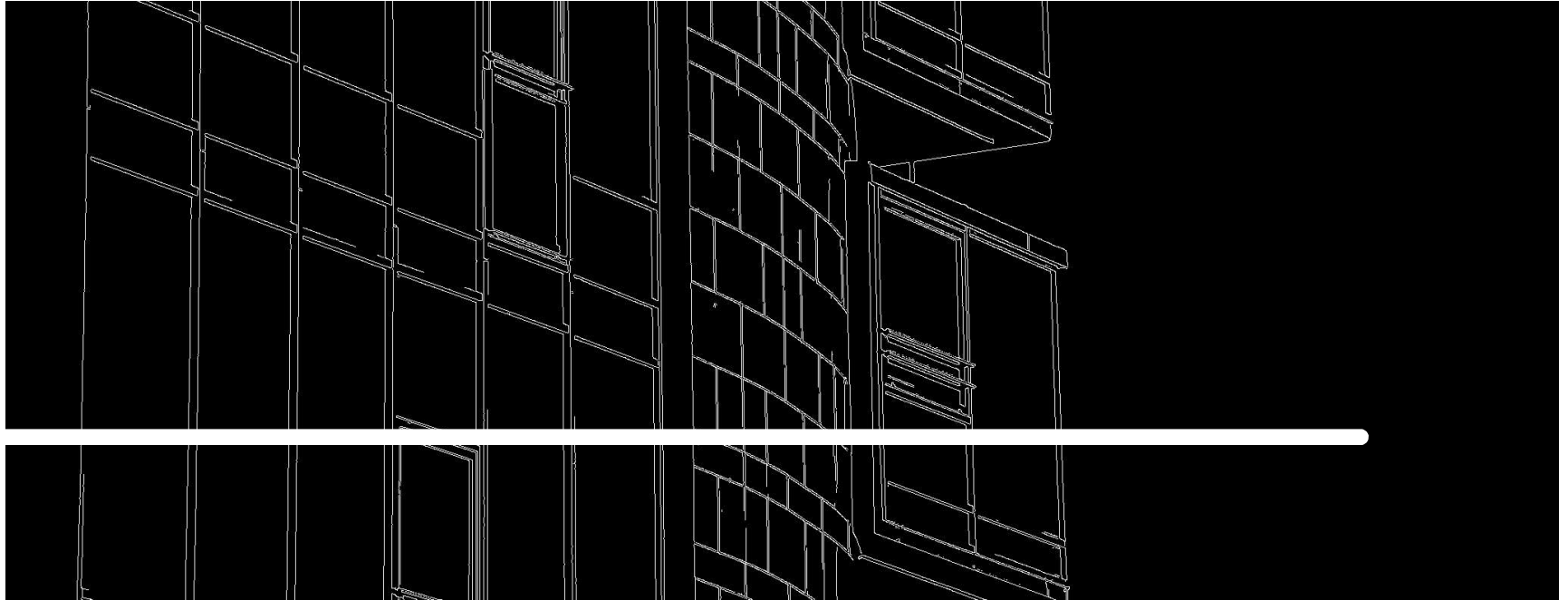
if img is None:
    print("Error: Image 'Ex4.jpg' not found. Please ensure the image is uploaded to the Colab environment (e.g., to the /cont
else:
    kernel = np.ones((5,5), np.uint8)

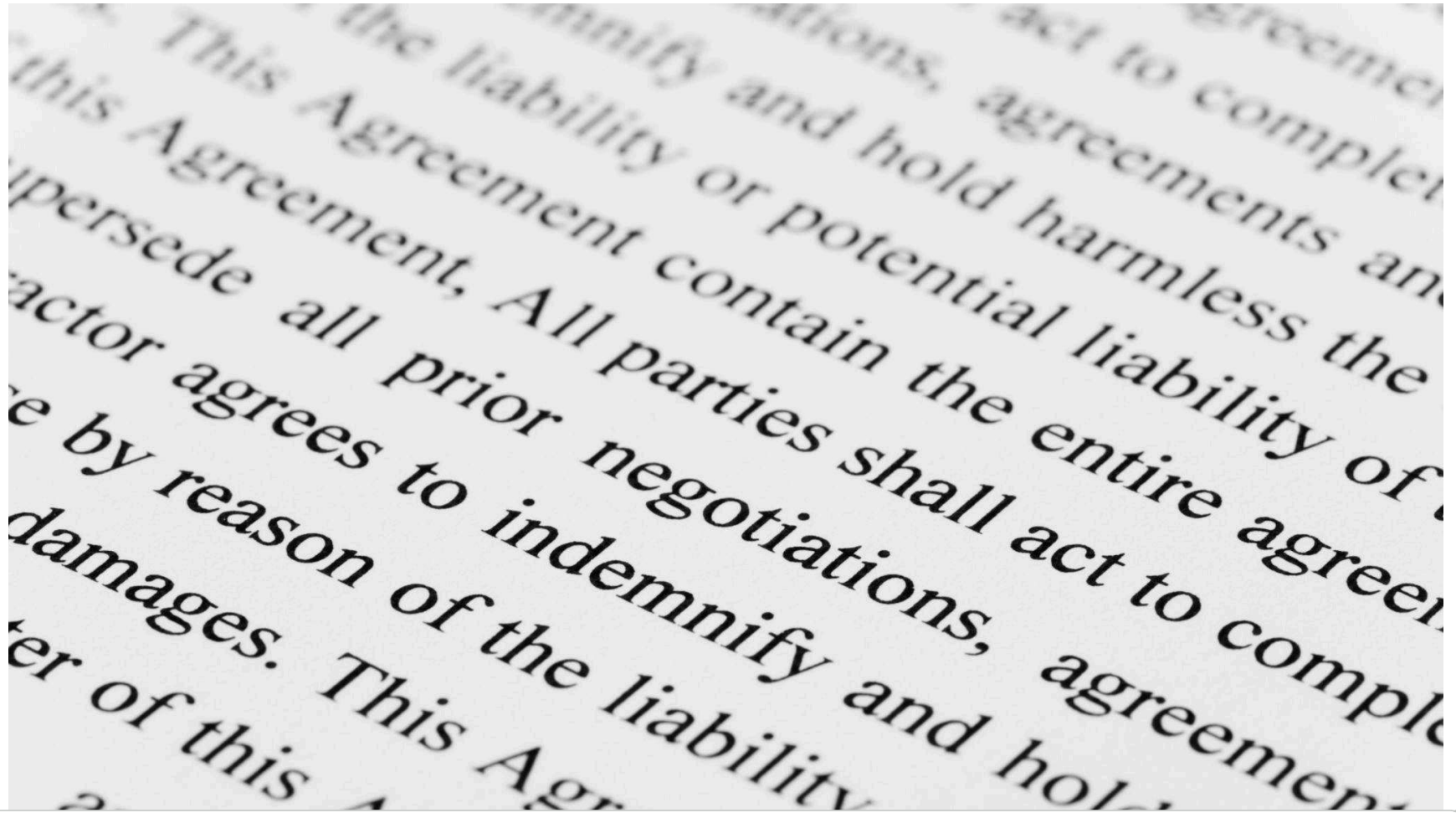
    dilated_img = cv2.dilate(img, kernel, iterations=1)

    cv2_imshow(img)
```

```
cv2.imshow('dilated_img')
```

```
cv2.imwrite("Dilated_Image.jpg", dilated_img)
```





```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow

img = cv2.imread("Ex5.jpg") # Corrected path for Colab environment

if img is None:
    print("Error: Image 'Ex5.jpg' not found. Please ensure the image is uploaded to the Colab environment (e.g., to the /content directory).")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5,5))

    eroded = cv2.erode(gray, kernel, iterations=1)
```

```
cv2_imshow(img)  
cv2_imshow(eroded)  
  
cv2.imwrite("eroded_image.jpg", eroded)
```





```
import cv2
from google.colab.patches import cv2_imshow

video_path = "Ex6.mp4"
cap = cv2.VideoCapture(video_path)

if not cap.isOpened():
    print(f"Error: Could not open video {video_path}. Please ensure the video is uploaded to the Colab environment.")
else:
    ret, frame = cap.read() # Read the first frame

    if ret:
        print("Displaying the first frame of the video:")
        slow = cv2.resize(frame, None, fx=0.5, fy=0.5)
```

```
fast = cv2.resize(frame, None, fx=2.0, fy=2.0)

print("Slow Motion (first frame):")
cv2_imshow(slow)
print("Fast Motion (first frame):")
cv2_imshow(fast)
else:
    print("Error: Could not read the first frame from the video.")

cap.release()
```



Displaying the first frame of the video:
Slow Motion (first frame):



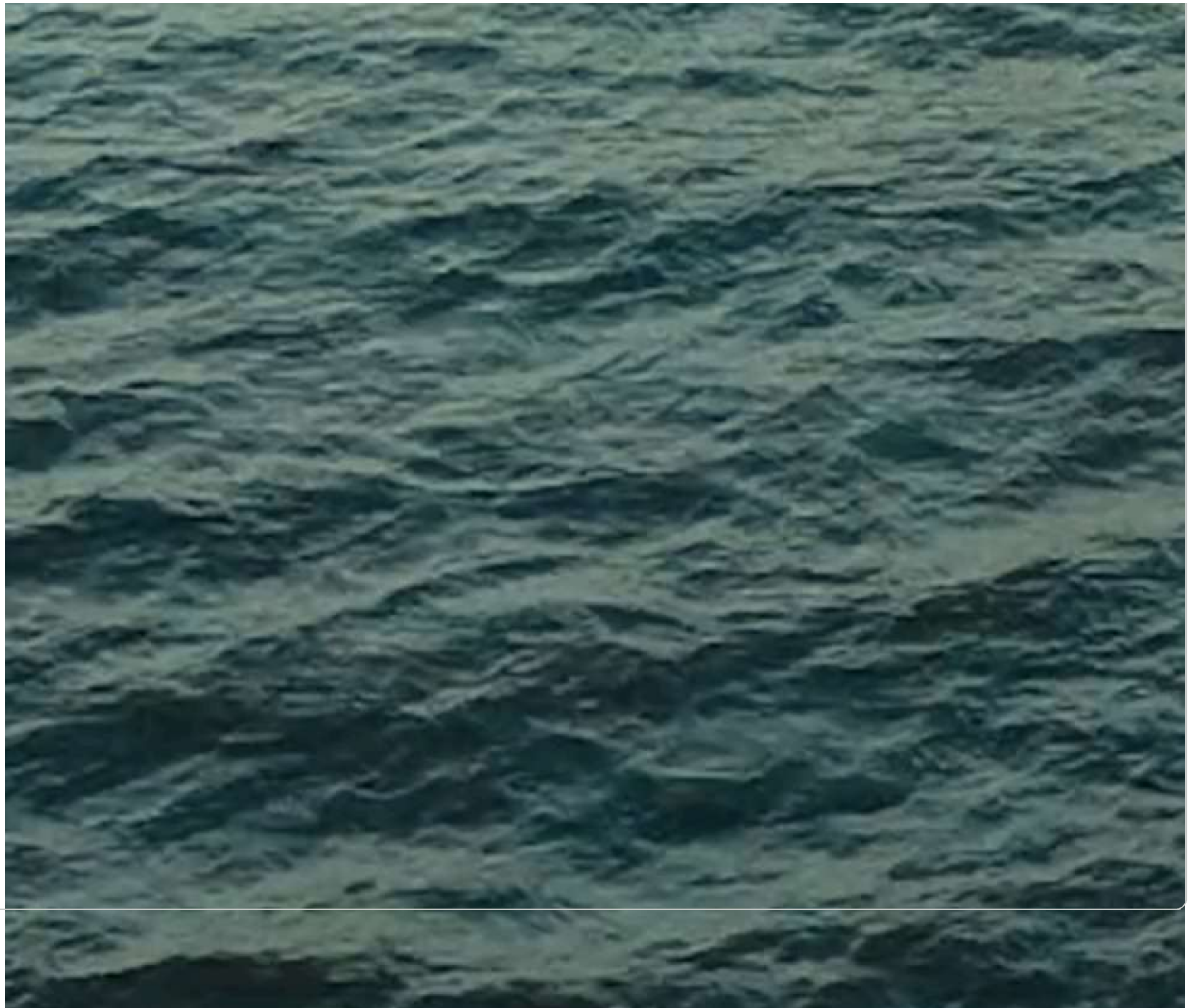


Fast Motion (first frame):









```
import cv2
from google.colab.patches import cv2_imshow

video_path = "Ex6.mp4" # Using an existing video file instead of webcam
cap = cv2.VideoCapture(video_path)

if not cap.isOpened():
    print(f"Error: Could not open video {video_path}. Please ensure the video is uploaded to the Colab environment.")
else:
    print(f"Processing first frame of video: {video_path}")
    ret, frame = cap.read() # Read the first frame

    if ret:
        slow = cv2.resize(frame, None, fx=0.5, fy=0.5)
        fast = cv2.resize(frame, None, fx=2.0, fy=2.0)

        print("Slow Motion (first frame):")
        cv2_imshow(slow)
        print("Fast Motion (first frame):")
        cv2_imshow(fast)
    else:
        print("Error: Could not read the first frame from the video.")

cap.release()
```



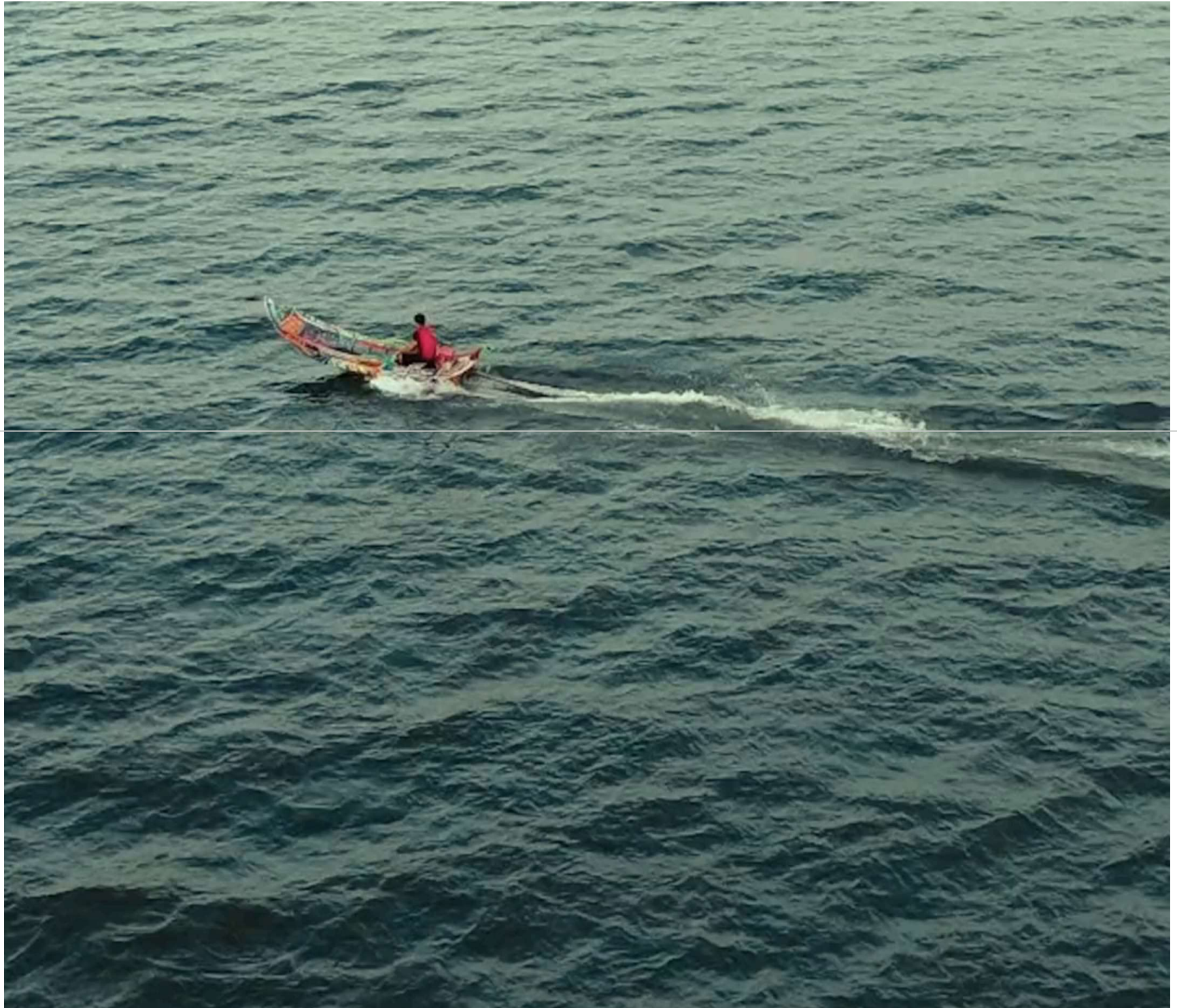
Processing first frame of video: Ex6.mp4
Slow Motion (first frame):





Fast Motion (first frame):





```
import cv2
from google.colab.patches import cv2_imshow # Import cv2_imshow for Colab compatibility

image_path = "Ex8.jpg" # Corrected path for Colab environment
image = cv2.imread(image_path)

if image is None:
    print(f"Error: Image '{image_path}' not found. Please ensure the image is uploaded to the Colab environment.")
else:
    height, width = image.shape[:2]

    bigger = cv2.resize(image, (int(width*3), int(height*3)))
    smaller = cv2.resize(image, (int(width*0.5), int(height*0.5)))

    print("Original Image:")
    cv2_imshow(image)
    print("Bigger Image:")
    cv2_imshow(bigger)
    print("Smaller Image:")
    cv2_imshow(smaller)

    cv2.imwrite("Bigger_Image.jpg", bigger)
    cv2.imwrite("Smaller_Image.jpg", smaller)

    # Removed cv2.waitKey(0) and cv2.destroyAllWindows() as they are not supported in Colab
```


Original Image:



Bigger Image:





Smaller Image:



```
import cv2
from google.colab.patches import cv2_imshow # Import cv2_imshow for Colab compatibility

image_path = "Ex9.jpg" # Corrected path for Colab environment
img = cv2.imread(image_path)

if img is None:
    print(f"Error: Image '{image_path}' not found. Please ensure the image is uploaded to the Colab environment.")
else:
    clockwise = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)

    counter = cv2.rotate(img, cv2.ROTATE_90_COUNTERCLOCKWISE)

    print("Original Image:")
    cv2_imshow(img)
    print("Clockwise Rotation:")
    cv2_imshow(clockwise)
    print("Counter Clockwise Rotation:")
    cv2_imshow(counter)

    cv2.imwrite("Clockwise.jpg", clockwise)
    cv2.imwrite("CounterClockwise.jpg", counter)

# Removed cv2.waitKey(0) and cv2.destroyAllWindows() as they are not supported in Colab
```



Original Image:



Clockwise Rotation:







Counter Clockwise Rotation:







```
import cv2
from google.colab.patches import cv2_imshow # Import cv2_imshow for Colab compatibility

image_path = "Ex10.jpg" # Corrected path for Colab environment
image = cv2.imread(image_path)

if image is None:
    print(f"Error: Image '{image_path}' not found. Please ensure the image is uploaded to the Colab environment.")
else:
    print("Original Image:")
    cv2_imshow(image)

# Removed cv2.moveWindow() as it is not supported in Colab
# Removed cv2.waitKey(0) and cv2.destroyAllWindows() as they are not supported in Colab
```

Original Image:



```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow # Import cv2_imshow for Colab compatibility

image_path = "Ex11.jpg" # Corrected path for Colab environment
img = cv2.imread(image_path)

if img is None:
    print(f"Error: Image '{image_path}' not found. Please ensure the image is uploaded to the Colab environment.")
```